



Pick and Summarize: Integrating Extractive and Abstractive Speech Summarization

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Abstract

Speech summarization condenses long speech while preserving essential content. Recently, there has been growing interest in end-to-end (E2E) abstractive speech summarization, which directly generates a text summary from spoken input. However, abstractive summarization of lengthy speech sequences presents challenges, such as identifying key information within very long speech. In this paper, we hypothesize that first addressing the simpler task of extractive summarization can help with these aforementioned long-sequence challenges and improve overall summarization performance. To this end, we introduce an extractive-abstractive summarization model that exploits auxiliary information from extractive summaries generated directly from raw speech input to enhance abstractive speech summarization. Experiments on a web presentation corpus demonstrate consistent gains with our proposed method, achieving up to 1.4-point gains in METEOR score over a strong abstractive summarization baseline.

Index Terms: speech summarization, abstractive, extractive, multi-task learning

1. Introduction

Speech summarization aims to condense very long speech input into a concise summary that preserves essential content and key highlights [1]. Summarization approaches are categorized into two styles [2, 3]: extractive and abstractive. Extractive summarization selects key utterances from a spoken document. It is particularly effective in structured domains, such as news broadcasts, where formal discourse and logical organization facilitate accurate sentence selection. Abstractive summarization generates text that captures the semantic essence of the spoken document. It excels in informal settings like conversations and meetings, where rephrasing disfluent speech is crucial. Consequently, abstractive speech summarization has recently received more interest than extractive speech summarization.

Many natural language processing (NLP) studies report the benefits of combining extractive and abstractive summarization task [4–6]. For example, sharing knowledge between extractive and abstractive summarization processes via pre-training or multi-task learning has been shown to enhance the performance of abstractive summarization [5, 7]. Another study [8] provides an extractive summary, instead of the whole document, to an abstractive summarization model that converts it to an abstractive summarization. However, these prior works were limited to text summarization. In this paper, we propose to extend the ideas of these works to end-to-end (E2E) speech summarization.

Recently, there has been an increased interest in E2E abstractive speech summarization, which directly generates a text summary from the input speech [9–15]. Such E2E models

promise several advantages over, e.g., cascade systems performing automatic speech recognition (ASR) followed by text summarization, with two independent models. First, E2E models can be robust to recognition errors [11]. Moreover, they can exploit paralinguistic information (e.g., emotion, emphasis), which was shown to be important for speech summarization [16]. However, current E2E abstractive speech summarization systems still sometimes lack behind cascade systems in some domains [11].

One reason could be the limited training data, but we believe another contributing factor is the challenge of abstracting very long speech sequences directly. In this paper, we hypothesize that first tackling the relatively easier task of extractive summarization could help mitigate the aforementioned long-sequence challenge in abstractive summarization. We propose E2E extractive-abstractive (ex-abstractive) summarization, a novel E2E abstractive speech summarization approach conditioned on extractive summaries. We introduce two implementations of the approach. First, we develop a *modular* approach, where the ex-abstractive summarization model accepts speech embeddings and an extractive summary obtained from an independent extractive model as inputs. As a side contribution, for this *modular* system, we develop a new extractive speech summarization model, which transcribes the important utterances directly instead of selecting them from ASR transcriptions as in prior works [17]. Second, we propose a *monolithic* approach, where one model sequentially generates the extractive followed by the abstractive summaries.

The proposed ex-abstractive summarization models preserve the advantages of E2E systems, i.e., being robust to ASR errors and capable of exploiting paralinguistic information. In addition, we expect that multi-task learning and conditioning the abstractive summarization on the extractive summaries will help find important contents from very long speech sequences.

We evaluate our framework on a web presentation corpus, which contains 664 hours of talks with human-annotated summaries. Experimental results demonstrate a 1.4 points improvement in METEOR scores [18] with our proposed abstractive summarization conditioned on extractive summaries compared with conventional systems [11]. Further analysis reveals that extractive summaries prevent the content from being lost in abstractive summarization.

2. Related works

Early works on speech summarization focused on extractive approaches. Prior works typically formulate extractive summarization as a binary classification task [19–21]. Each sentence in transcriptions is assigned a label (0, 1) to indicate its exclusion (0) or inclusion (1) in summary. The extractive summary comprises all sentences with the inclusion label 1. This ap-

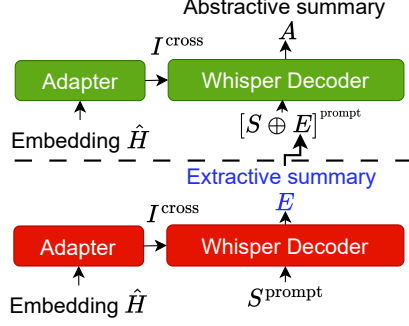


Figure 1: *Modular system*

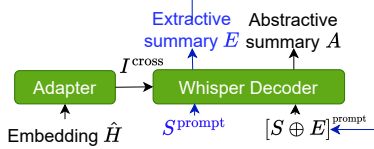


Figure 2: *Monolithic system*

proach ensures the correctness of the content at the sentence level. There have been many approaches that select important sentences from transcriptions using Bayesian classifier, support vector machine, and neural networks [17, 22–24]. More recently, EESUM [21] performs E2E speech-to-speech extractive summarization directly, without transcription. It relies on term frequency–inverse document frequency scoring over self-supervised speech representations (e.g., Wav2Vec [25]) to select salient content in the spoken document. This work is out of our scope because we focus on generating text summaries instead of speech summaries.

With the advent of sequential models [26], abstractive speech summarization has become pre-dominant [7, 27, 28] since rephrasing disfluent speech is crucial, it is often more appropriate for speech summarization. In this work, unlike prior abstractive speech summarization approaches, we condition the processing on extractive summaries. For this, we design a novel extractive speech summarization model, which transcribes only important utterances from the spoken document instead of selecting them from the transcriptions.

3. End-to-end speech summarization

3.1. Abstractive speech summarization

The E2E speech summarization system receives an input speech signal X and generates an abstractive summary A . Previous studies have fine-tuned pre-trained ASR models for speech summarization tasks [11, 29]. Here, we use the Whisper [30] encoder and decoder but introduce two modifications to address the long-sequence and speech summarization tasks.

First, we run the encoder in a chunk-wise manner since Whisper automatically segments long speech into 30-second chunks and applies padding to the final chunk during preprocessing. The encoded speech sequence is obtained as follows:

$$H = \text{ENC}(X; \theta_e), \quad (1)$$

where, $\text{ENC}(\cdot)$ denotes the Whisper encoder function with parameters θ_e . $H \in \mathbb{R}^{C \times L \times D}$ is the sequence of Whisper embeddings, C is the number of chunks, L is the length of each speech chunk, and D is the embedding dimension. Since each speech chunk is encoded independently, the model cannot capture long-range contextual information across chunks. To ad-

dress this, we concatenate the embeddings along the C dimension, obtaining $\hat{H} \in \mathbb{R}^{(CL) \times D}$.

To further enhance long-context learning for summarization, we introduce an adapter network between the encoder and the decoder that processes $\hat{H}$ as follows

$$I = \text{ADA}(\hat{H}; \theta_a) \in \mathbb{R}^{(CL) \times D}, \quad (2)$$

where, the $\text{ADA}(\cdot)$ is a two-layer Transformer network [31], the θ_a denotes its parameters, and the I is the final embedding sequence of the encoding process.

For the decoder, we leverage the prompt framework of Whisper to control target generation. The decoder receives I and special tokens S , and generates the target sequence A in an auto-regressive manner as follows:

$$a_k = \text{DEC}(a_{1:k-1}, I^{\text{cross}}, S^{\text{prompt}}; \theta_d), \quad (3)$$

where a_k is a k -th token of A , S^{prompt} denotes that the special tokens S is inserted to the Whisper decoder as a prompt. I^{cross} denotes the speech embedding imputing cross-attention function. $\text{DEC}(\cdot)$ is the Whisper decoder function, and θ_d denotes its parameters. We extend the Whisper’s special token format as

$$S = [\langle \text{startoftranscript} \rangle \oplus \langle \text{en} \rangle \oplus t \oplus \langle \text{notimestamp} \rangle], \quad (4)$$

where $\langle \text{startoftranscript} \rangle$ is the start of sentence token, $\langle \text{en} \rangle$ denotes the language (English) ID, $\langle \text{notimestamp} \rangle$ denotes not generating timestamp. Additionally, we introduce $t \in \{\langle \text{extract} \rangle, \langle \text{abstract} \rangle\}$ as a summarization task,

The system is trained using the cross-entropy loss between the ground-truth summary and the prediction. During training, we freeze θ_e (encoder parameters) and fine-tune the θ_a (adapter parameters) and θ_d (decoder parameters).

3.2. Extractive speech summarization

In this paper, we aim to condition the decoding process of the abstractive summarization on the extractive summary. For this purpose, it is more convenient to input the text token of the extractive summary instead of utterance-level exclusion/inclusion labels because the decoder is pre-trained with text sequences. Therefore, we reformulate extractive summarization as a sequence generation task. Given an input speech signal X , an extractive summarization model generates the target extractive summary E directly.

The extractive summary E is a subset of the original spoken sentences containing as much content as possible. If the training corpus provides extractive summaries, we can directly use these for training. If not, we can create pseudo extractive summaries for training targets using automatic summarization evaluation metrics [7] and search algorithm. We followed previous research and employed BERTScore [32] as automatic summarization metrics. We calculated BERTScore for each sentence of a ground-truth transcription and a human abstractive summary. We used a greedy search algorithm to generate a pseudo extractive summary, which maximizes the BERTScore. Extractive summaries are generated automatically by selecting U sentences from the transcription that maximize the BERTScore for the reference abstractive summary. The number of extracted sentences, U , is a hyperparameter. The summarization process is thus identical to that of abstractive summarization described in Section 3.1, only the training target differs. However, note that this extractive summarization is relatively easier than abstractive summarization, as its task is merely to conduct long-form ASR with a limited number of sentences.

4. Proposed ex-abstractive

We propose two frameworks to integrate extractive and abstractive summarization during training and inference. We introduce the *modular* and *monolithic* ex-abstractive approaches shown in Figs. 1 and 2, respectively.

4.1. Modular system

In the *modular* system, as illustrated in Fig. 1, we treat that extractive and abstractive summarization are separately conducted with two different modules: one for extractive summarization and another for ex-abstractive summarization.

The extractive summary E is obtained by training a separate Whisper decoder, where A is replaced with E in Eq. (3), as described in Sec. 3.2. The ex-abstractive model takes both the input speech X and the generated extractive summary E as input. The encoder generates I similar to Eqs. (1) and (2). Unlike Eq. (3), the decoder is conditioned on the extractive summary E as an additional prompt to generate the abstractive summary A as:

$$a_k = \text{DEC}(a_{1:k-1}, I^{\text{cross}}, [S \oplus E]^{\text{prompt}}; \theta_d). \quad (5)$$

Therefore, A is generated based on both I and E .

The *modular* system’s loss is the same as that described in Sec. 3.1. The *modular* system only trains for the abstractive summary tasks, so we ignore the prompt composed of the special tokens and the extractive summary at loss calculation. In the *modular* system, the ex-abstractive summarization model does not share the knowledge for solving extractive tasks.

4.2. Monolithic approach

Next, we propose a *monolithic* approach and try to share knowledge between extractive and abstractive summarization tasks in a single model. Unlike the *modular* system, the *monolithic* approach generates sequentially extractive and abstractive summaries using the same model and a single decoding process as shown in Fig 2. We implement this model by simply changing the target sequence in Eq. (3) to $Y = [E \oplus \langle \text{abstract} \rangle] \oplus A$ in auto-regression decoding, where $\langle \text{abstract} \rangle$ works as an end-of-sequence token for E . The decoder generates Y as follows;

$$y_k = \text{DEC}(y_{1:k-1}, I^{\text{cross}}, S^{\text{prompt}}; \theta_d). \quad (6)$$

Here, S^{prompt} is as defined in Eq. (4), with t set to $\langle \text{extract} \rangle$, y_k is the k -th token of Y , $y_{1:k-1}$ is the sequence of previously generated tokens. Consequently, like the *modular* system, the abstractive summary generation is conditioned on the extractive summary.

The training loss is the sum of the cross entropy loss for the extractive and abstractive summary, each normalized by the sequence length. This normalization ensures that multi-tasks of varying lengths are balanced in the total loss calculation, preventing the extractive summarization from generating excessively long sequences based on the input X [33].

5. Experiments

5.1. Experimental settings

We conduct all experiments within the ESPnet toolkit [34]. We evaluated our proposed method using a web presentation dataset, which consists of 3384 talks, 664 hours of speech and human-annotated transcriptions and abstractive summaries. The evaluation set contains 424 talks corresponding to a total of 84 hours of speech.

For the initialization of E2E systems, we employed the Whisper-large-v2 model with 32 encoder and decoder layers. First, we fine-tuned the Whisper model for long-form speech recognition task. We compared our approach against conventional E2E and cascade speech summarization models [11]. The cascade system, following previous study [11], uses Whisper-base to generate transcriptions and T5 [35] for text summarization. The E2E baseline system consists of the Whisper model fine-tuned to generate abstractive summaries as in Sec. 3.1, but without our extension to handle long recordings. Therefore, it can only accept the first 30 sec of the speech. We name this model as WhisperSUM.

In this work, following Sec. 3.1, we also prepared an E2E abstractive summarization system for long-form speech (maximum duration of training data is 20 minutes). Because of memory and computing limitations, we cannot fine-tune the Whisper encoder parameters of the E2E model. However, it can consider a longer context than WhisperSUM through the adapter network and the decoder.

To construct the *modular* system, we generated two separate models (an extractive and an ex-abstractive one), by fine-tuning the pre-trained long-form ASR model separately for extractive and ex-abstractive summarization. We use ground-truth extractive summaries to condition abstractive summarization when training the ex-abstractive model. In contrast, for the *monolithic* system, we fine-tuned the pre-trained long-form ASR model directly for ex-abstractive summarization to output extractive and abstractive summaries as explained in Sec. 4.2.

The length of extractive summaries is a hyper-parameter affecting the performance of the ex-abstractive summarization model. Based on our analysis described in Sec. 5.3, we set the length of extractive summaries to 10% of the number of sentences in the transcriptions. Note that the extractive summary length is controlled as a hyper-parameter in the data preparation stage. The *modular* system’s extractive summarization model and *monolithic* model are trained with a specific summarization length for the references. At inference time, they perform autoregressive decoding until the end of the sequence token appears.

We trained all models with a batch size of 1 and validated every 1,000 steps. Early stopping was applied with a patience of 10, and models were fine-tuned for up to 1M iteration. We used a ReduceLROnPlateau scheduler¹ with an initial learning rate of $1e-5$, and optimized using AdamW [36] with $\beta = 0.9$ and $\beta_2 = 0.999$. For the abstractive and ex-abstractive models, validation was based on abstractive summarization accuracy, while the extractive summarization model in the *modular* system was validated based on extractive summarization accuracy. We evaluated summarization performance using BERTScore (BERT) [32], METEOR (MTR) [18], and ROUGE-Longest (R-L) [37].

5.2. Summarization results

Table 2 compares performances of different systems using human abstractive summaries as references (top) and automatically generated pseudo extractive summaries as references (bottom), as introduced in Sec. 3.2. We evaluate the baseline systems only in terms of abstractive summaries, as they do not generate extractive summaries. In contrast, our proposed *modular* (system No. 4) and *monolithic* (system No. 5) systems can generate both extractive (No. 4-1,5-1) and abstractive (No. 4-2,5-2) summaries, and we evaluate the performance for both

¹https://pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.ReduceLROnPlateau.html

Table 1: Summarization example

Ground-truth extractive	So I began to wonder what it might be like to parent my children in a war zone and a refugee camp . So the question I had was could parent training programs be useful for families while they were still in ...
Ground-truth abstractive	What it's like to be a parent in a war zone & How do parents protect their children and help them feel secure again when their homes are ripped apart by war ? In this warm-hearted talk , psychologist Aala El-Khani shares her work supporting – and learning from – refugee families affected by the civil war in Syria . She asks: How can we help these loving parents give their kids the warm , secure parenting they most need ?
Monolithic extractive	So I began to wonder what it might be like to parent my children in a war zone and a refugee camp . So the question I had was could parent training programs be useful for families while they were still in ...
Monolithic abstractive	How we can help refugees children in Syria & Refugee camp teachers are the most innocent and vulnerable victims of conflict, but they are also the most vulnerable. How can we help refugees in Syria in general? In this inspiring talk, psychologist and parent trainer Hala Al-Khalili explains how we can help refugees .
End-to-end abstractive	How to help children in conflict, war , chaos and conflict & Refugee camp teachers are the most innocent and vulnerable victims of war and conflict. In this inspiring talk, psychologist and TED Fellow Zahra Al-Khalabi explains how we can help children in conflict, war, chaos and conflict.

Table 2: Abstractive and Extractive summarization performance. $system^E$ and $system^A$ output extractive and abstractive summaries, respectively. The conventional systems (1,2,3) only perform abstractive summarization.

No.	System	R-L	MTR	BERT
Evaluation for human abstractive reference				
1	Cascade [11]	18.2	12.6	87.1
2	WhisperSUM [11]	16.8	11.9	85.1
3	End-to-end	16.8 $\pm$ 1.1	11.5 $\pm$ 0.9	85.1 $\pm$ 1.0
4-1	Modular ^E	8.3 $\pm$ 0.8	10.4 $\pm$ 1.0	70.5 $\pm$ 1.2
4-2	Modular ^A	18.5 $\pm$ 1.1	13.6 $\pm$ 1.2	87.6 $\pm$ 1.3
5-1	Monolithic ^E	9.0 $\pm$ 0.8	11.7 $\pm$ 1.0	73.1 $\pm$ 1.1
5-2	Monolithic ^A	18.8 $\pm$ 1.2	14.0 $\pm$ 1.2	87.8 $\pm$ 1.1
Evaluation for pseudo extractive reference				
4-1	Modular ^E	50.9 $\pm$ 1.1	40.6 $\pm$ 1.0	93.6 $\pm$ 0.9
5-1	Monolithic ^E	43.2 $\pm$ 1.0	40.1 $\pm$ 1.2	90.0 $\pm$ 0.9

summary styles. Our proposed *modular* (No. 4-2) and *monolithic* (No. 5-2) systems outperformed the baseline end-to-end and cascade models, with the *monolithic* system achieving the highest performance in all three metrics. The provided extractive summary includes topic words. These topic words support abstractive summarization finding the important content from long speech sequence, as described in Sec 5.4.

Due to the gap between written and spoken style, the extractive summaries (spoken style) achieve lower scores than the abstractive ones (written style) when evaluated with human abstractive summary references (written style). Therefore, to confirm if the generated summaries are effectively extractive, we measure how similar they are to the reference pseudo-extractive summaries, obtained as explained in Sec. 3.1. The bottom two rows of Table 2 evaluate the quality of the extractive summaries generated with the *modular* and *monolithic* systems.

The results show that both *modular* and *monolithic* systems generate extractive summaries very close to the pseudo ones based on all three metrics. Since the extractive summarization model in the *modular* system is explicitly trained for extractive task, it achieves the best performance on extractive targets. Although the extractive summaries produced by the *monolithic* system may deviate from reference extractive summaries, they ultimately enhance the final abstractive summarization perfor-

Table 3: Monolithic system’s abstractive summarization results for each extractive summarization ratio. The results are obtained using ground-truth extractive summary at inference.

Extracted sentence	RL	MTR	BS
0	16.8	11.5	85.1
2	18.2	13.4	86.8
4	18.8	13.3	86.8
6	18.6	13.8	87.6
8	19.0	13.9	87.5
10	19.0	13.8	87.6

mance.

5.3. Optimal summarization ratio of extractive summary

In this section, we explore the optimal value of U introduced in Sec. 3.2. We set $U = rZ$ where r is the summarization ratio and Z is the number of sentences in the input speech. During extractive data creation, we vary r and train separate *monolithic* systems for each. Table 3 presents the abstractive summarization performance when using ground-truth extractive summaries as input. These results indicate that the extractive summarization ratio of 5–10% is optimal for abstractive summarization, yielding the best score in the *monolithic* system.

5.4. Examples

Table 1 shows the summarization example of *monolithic* and E2E systems. This example shows the first several sentences of *monolithic* system and ground-truth extractive summary. The extracted utterances include topic words (highlighted by red, blue, and green) that appear in the ground-truth abstractive summary. The summary generated with the *monolithic* system contains two topic words (**giver**, **taker**) and lost **matcher**. However, the E2E system’s summary only contains one topic word (**giver**). This result shows an example where extractive summarization can pick topic words from utterances and prevent the content from being lost in abstractive summarization.

6. Conclusion

We propose an E2E speech summarization system that conditions extractive summary to abstractive summarization. Extractive summaries prevents the content from being lost in abstractive summarization and improves summarization performance. In future works, we will apply our proposal to a wide domain.

7. References

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